

Accelerating the Science-to-Technology Transition: How Generative AI Shapes the Speed and Breakthroughness of Corporate Innovation

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Abstract

Corporate R&D draws on academic literature, yet the transition from published science to patented invention has historically taken years. Did the post-2023 adoption of generative AI (GenAI) shorten the lag between academic publication and corporate patent citation, and does GenAI-era R&D reorient what science patents build on? This extended abstract reports preliminary results from an ongoing study of these questions. Using 429,111 patent–paper citation links from the Lens covering US filings from 2018 to 2025, we estimate a difference-in-differences design: technology classes with high GenAI exposure (CPC G06N, G16B, G16C, C40B) form the treated group, mature mechanical classes (F16B, F16H, B65D) form the control group, and the treatment period begins in 2023. We report three preliminary findings. First, the citation-weighted lag between a cited paper’s publication and the citing patent’s filing *lengthened* in GenAI-exposed classes relative to controls after 2023 (about +0.20 log points, robust across progressively stricter truncation guards), while the median lag of the typical patent was unchanged: the effect is a composition effect, not a slowdown of the average patent. Second, the composition margin moved: the share of citations to papers three years old or younger fell from 19.7% to 11.9% in treated classes (a patent-level difference-in-differences of –4.9 percentage points), and the citation mass moved into the four-to-eight-year window (2015–2019 publications for 2023–2024 filings), not into the oldest stock. Third, a placebo estimation with a false 2017 treatment on 2015–2019 data shows no lag effect, while fresh-science intake in treated classes was *rising* before 2023: the decline is a reversal, but pre-trends were not parallel. Two readings remain open and are tested next: a shift toward transformer-era classics, or slower indexing of the newest papers. Contrary to the acceleration narrative, the first patent-record evidence shows no speed-up in the uptake of new science in GenAI-exposed classes.

Keywords

generative AI, science of science, science-to-technology lag, patents, breakthrough innovation, difference-in-differences, semantic homogenization

1. Introduction and Motivation

Corporate R&D depends on academic science: firms read the literature, absorb new findings, and translate them into technologies that surface as patents. This translation is slow. Prior work has documented a science-to-technology lag that typically spans years between the publication of a scientific result and its citation in a patent [1, 2]. In biomedicine, patent citations trail scientific citations by a median of six years [7], and the time to first patent citation averages roughly a decade, varying with basicness and novelty [8].

After 2023, generative AI (GenAI) gave R&D teams tools to synthesize scientific knowledge, automate code, and screen literature at scale. Whether this improves corporate innovation is open: GenAI may help firms connect distant fields, or rivals relying on the same foundation models may converge on faster but less distinctive, less disruptive patents [3, 4].



2. Research Questions

- **RQ1 (Speed).** Has GenAI adoption after 2023 shortened the lag between academic publication and corporate patent citation?
- **RQ2 (Homogenization, descriptive).** Do patent abstracts in GenAI-exposed classes show semantic convergence across firms after 2023?

3. Data and Methods

Data. We draw patent records and their citations to scholarly works from the Lens,¹ which links patents to the papers they cite with dated records on both sides, complemented by OpenAlex for paper-level attributes [5]. The sample covers filings from 2018 through 2025 in seven CPC classes selected from the official USPTO/EPO definitions: a GenAI-exposed treated group (G06N, artificial intelligence and machine learning; G16B, bioinformatics; G16C, computational chemistry and materials; C40B, combinatorial chemistry) and a low-exposure mechanical control group (F16B, fastening devices; F16H, gearing; B65D, containers and packaging).

Identification. Difference-in-differences on the citation links: the log lag (days from paper publication to patent filing) and the patent-level share of citations to papers ≤ 3 years old are regressed on treated class $\times$ post-2023 filing, with class and filing-year fixed effects and standard errors clustered by class-year. Field-wide shocks common to both groups, such as examination speed-ups or drifting citation practices, are absorbed.

Results (preliminary). The treated $\times$ post coefficient on the pair-level log lag is +0.20 ($p < 0.001$) under a uniform ten-year citation window, and strengthens rather than attenuates as truncation guards tighten; at the patent level the median-lag effect is null, locating the change in citation composition rather than in the typical patent’s speed. The patent-level share of fresh science falls by 4.9 percentage points in treated classes ($p = 0.001$; from 19.7% to 11.9%), robust to restricting filings to mid-2024. Shifted windows locate where the mass went: the four-to-eight-year share rises (+8.2 points, $p = 0.02$) while the share of papers eight years or older does not (−5.3 points, n.s.): the shift is toward 2015–2019 publications, not the oldest stock. This fits a move toward transformer-era classics and also slower resolution of the newest papers; Lens resolution rates show no post-2023 break in treated classes, which weakens but does not exclude the second reading. Paper-level controls [8] and an indexing-independent citation source are being added next.

Robustness. A placebo estimation on 2015–2019 with a false 2017 treatment date yields no lag effect ($p = 0.46$). On the composition margin the fresh-science share was rising in treated classes before 2023, which rules out field maturation but means pre-trends were not parallel; an event-study with yearly leads precedes any causal reading. For RQ2, VOSviewer term maps [9] of G06N abstracts before and after 2023 show a vocabulary shift toward generative-model terms but no collapse into fewer clusters. We acknowledge the limitations of direct citation linkage [10]; we retain it because our question is temporal and direct citations are the only linkage yielding two dated events.

4. Contributions

For corporate strategy, GenAI-era invention shows no acceleration in fresh-science uptake so far, so absorptive capacity in frontier literature remains a differentiator. For the science of science, the results link the GenAI shock to the debate on declining disruptiveness [4]: a reallocation of patent citations away from the newest science, its mechanism still open. Archive:²

¹<https://www.lens.org>

²<https://anonymous.4open.science/r/sci-tech-lag-C45D/>

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